

Amir Masoud Aghaei

AI\ML ENGINEER · PYTHON

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Personal Profile

Curious and driven Artificial Intelligence Master's student at the University of Bologna, targeting an AI/ML Engineer position. Skilled in Python, Machine Learning (Scikit-Learn, PyTorch, Pandas, Numpy, Matplotlib), Computer Vision (OpenCV) and modern LLM architectures (RAG, LangChain, LangGraph, Vector Databases) with a knowledge in cloud environments like AWS.

Education

Università di Bologna

Master in Artificial Intelligence

- CGPA: 26.9/30 (so far)

Bologna, Italy

2024 - Present

University of Tehran

B.Sc. in Computer Science

- CGPA: 3.14/4

Tehran, Iran

2018 - 2023

Skills

Languages

Python, C++

ML / DL

PyTorch, TensorFlow, Scikit-Learn, HuggingFace, Transformers, Tokenizers, Embedding Models, Whisper

GenAI / LLMs

LangChain, LangGraph, RAG, OpenAI API, Vector Databases, Prompt Engineering

MLOps & Tools

Git, Linux, Docker, Streamlit, $\LaTeX$

Web

HTML5, CSS, WordPress, Jet Engine, Elementor

Human Languages

Persian (Native) · English (TOEFL 97: R26 L27 W22 S22) · Italiano (A2.1)

Notable Projects

Explainable EctHR Legal Outcome Prediction

transformers, pytorch, scikit-learn,

AWS SageMaker

NLP · Argument Mining · Multi-label Classification · Explainability

2026

- Two-stage pipeline: fine-tuned LegalBERT for premise extraction (F1: 0.96), then trained SVM/DT/EBM classifiers for multi-label violation prediction across 10 articles
- Improved extraction via similarity-filtered negatives, dynamic top-k selection, and per-article threshold tuning
- Achieved Macro F1: 0.49, Micro F1: 0.56 with concatenated pooling (2304-d) + SVM; explainability via weight vector attribution
- Deployed on AWS SageMaker; premise extraction improved F1 by 0.05 over full-text baseline

Sexism Detection on Social Media (EXIST Dataset)

transformers, tensorflow, pytorch,

scikit-learn, nltk

NLP · Text Classification · Hate Speech Detection · Fine-tuning Transformers

2026

- 4-class sexism classification on 3,300 English tweets using majority-voting label aggregation and robust preprocessing pipeline
- Compared BiLSTM with 100-d GloVe embeddings against fine-tuned Transformers (Twitter-RoBERTa, BERTweet) for domain-specific detection
- Achieved Macro F1: 0.50, Accuracy: 75.36% with Twitter-RoBERTa, outperforming BiLSTM (F1: 0.38) and BERTweet (F1: 0.45)
- Identified 11.8% accuracy drop on OOV samples; class imbalance caused minority class F1 < 0.30 requiring augmentation

Book Instance Detection on Shelves

OpenCV, NumPy, SIFT, FLANN,

RANSAC

Computer Vision · Feature Matching · Object Detection

2025

- Built a multi-instance book detection pipeline using SIFT keypoints, FLANN matching, and RANSAC homography estimation
- Applied iterative masked detection to locate multiple copies of the same book per scene, validated by quadrilateral geometry checks
- Enhanced keypoint detection under varying lighting via CLAHE contrast normalization
- Reported per-instance bounding boxes, corner coordinates, and pixel area for each detected book

Pet Breed Classification

PyTorch, torchvision, ResNet-18,

Oxford-IIIT Pet

Computer Vision · Deep Learning · Transfer Learning · CNN

2025

- Designed a custom 4-block CNN (PetNet) from scratch, reaching 63% test accuracy on 37-class pet breed classification
- Conducted ablation study comparing full vs. reduced-depth architectures to validate design choices
- Fine-tuned ResNet-18 with progressive layer unfreezing (epoch 26), AdamW, cosine annealing, and dropout — achieving 72% test accuracy
- Analyzed misclassified samples to identify data-driven bottlenecks: occlusion, inter-class similarity, and image quality

Sports Tournament Scheduling

MiniZinc, Z3, PuLP, Gecode, Chuffed,
CBC
2025

Combinatorial Optimization · CP · SAT · SMT · MIP

- Modeled NP-complete round-robin tournament scheduling under four paradigms: CP (MiniZinc), SAT, SMT, and MIP
- Designed symmetry-breaking and redundant propagation strategies; SAT with Z3 solved all instances up to $n = 20$ within 300s
- Benchmarked Gecode vs. Chuffed: Chuffed with symmetry-breaking reached $n = 16$; Luby restarts extended Gecode's reach to $n = 12$
- MIP with symmetry-breaking solved $n = 14$ in under 15s vs. timeout without it; all models unified under a single Docker interface

Urban Transit Network Analysis: Milan & Barcelona

NetworkX, pandas, geopandas,
folium, GTFS
2025

Network Science · Graph Analysis · Urban Computing · SNA

- Modeled Milan (4,913 stops) and Barcelona (2,771 stops) transit networks and computed 12 SNA metrics across different aspects
- Barcelona outperforms Milan in global efficiency (0.040 vs 0.030) and average path length (31 vs 40 stops) despite being the smaller network
- High modularity (Q: 0.89/0.93) and positive assortativity in both cities confirm a resilient rich-club hub topology
- Milan degrades faster under betweenness attacks (AUC 0.163 vs 0.202), exposing its radial structure's vulnerability

Certificates

The AI Engineer Course: Complete AI Engineer Bootcamp

Udemy
2026

365 Careers

- AI principles and practical applications
- Natural Language Processing in Python
- Large Language Models and Transformers
- Developing apps with orchestration tools like LangChain and LangGraph
- Retrieval-Augmented Generation Applications
- Vector databases using PineCone
- Creating AI-driven applications

Work Experience

ApplyClub

Remote

Web Designer

2024-2025

- Built and managed applyclub.com using WordPress, focusing on high performance and scalability.
- Developed complex features (Faculty Finder, Fundraising) using JetEngine for dynamic listings and advanced filtering.
- Enhanced UI/UX by implementing tailored HTML5, CSS3, and PHP code to extend plugin functionalities.
- Designed professional brand assets (Photoshop/Illustrator) and managed video content production.

Mehraein Mokhberi Consulting Firm

Tehran, Iran

Web Designer & CRM Manager

2023 - 2024

- Designed and developed company website utilizing Elementor page builder, CSS, HTML5, and Adobe Photoshop for graphic content.
- Successfully designed the drmokhberi.com website, enhancing its user experience and visual appeal.
- Managed CRM operations, ensuring effective communication and relationships with clients.